SSAU-Net: A Spectral–Spatial Attention-Based U-Net for Hyperspectral Image Fusion

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***Abstract*— Compared with the traditional remote sensing image, there is a large amount of spectral information in the hyperspectral image (HSI), which makes HSI better reflect the actual condition of surface features. However, due to the limitations of imaging conditions, HSI tends to have a lower spatial resolution. In order to overcome this issue, we propose a spectral–spatial attention-based U-Net named SSAU-Net for HSI and multispectral image (MSI) fusion. The SSAU-Net constructs a spectral–spatial attention module by a coordinate-attention (CA) module and an efficient pyramid split attention (EPSA) module, which can enhance the image's spectral information and spatial information. Meanwhile, the proposed network fully extracts the shallow and deep features of the images and finally generates high-resolution (HR) HSIs. Compared with the state-of-the-art HSI-MSI fusion methods, the experimental results verify that the proposed method has a better subjective and objective fusion effect.**

***Index Terms*— Deep learning, hyperspectral image (HSI), image fusion, multispectral image (MSI), spectral–spatial attention.**

I. Introduction

DUE to the rich spatial and spectral information, high-resolution hyperspectral image (HR-HSI) is often used in the tasks of target classification, target detection, and anomaly detection [1], [2], [3], [4], [5]. Attribute to the restriction of physical conditions of imaging equipment, it is an enormous challenge to improve the signal-to-noise ratio of HSIs due to the insufficient light energy in the case of dense image bands, which leads to a lower spatial resolution of HSI with more spectral bands. This seriously influences the application of HSI in computer vision-related tasks. With the advancement of remote sensing and the competition among various countries in the field of air and space, how to obtain high-quality HR-HSI is a focused issue in image processing. There are usually two ideas to boost the spatial resolution of low-resolution hyperspectral images (LR-HSI). One is the super-resolution of a single HSI. The other is the fusion of the high spatial resolution multispectral image (HR-MSI) and the LR-HSI. In this article, we prefer the latter as the HSI-MSI fusion method. Compared with the single-image super-resolution, an HSI-MSI fusion method with the auxiliary image can obtain a higher quality HR-HSI [6], [7], [8]. For example, pan-sharpening refers to obtaining HR-HSI by extracting spatial details from MSI and injecting them into HSI. Matrix decomposition-based HSI-MSI fusion methods include low-rank representation-based and sparse representation-based. For example, a coupled nonnegative matrix factorization (CNMF) algorithm was introduced by Fu et al. [7] for HSI fusion. It extracted the initialized end-member features in LR-HSI by using the vertex component analysis. Then, it combined matrix decomposition and a sensor sensing model to iteratively solve the abundance matrix of HR-MSI and the end-member matrix of LR-HSI. The result of the multiplication of the abundance matrix and the end-member matrix is the fusion result with HR. Yokoya et al. [8] used a linear mixing model to transform the fusion problem into the decomposition of end-member and abundance of two input images, which greatly improved the HSI-MSI fusion effect. Dian and Li [10] proposed an HSI-MSI fusion method based on low tensor multirank regularization. It made full use of spectral correlation and nonlocal similarity in HR-HSI and improved the fusion effect. Despite the considerable fusion results realized by the mentioned algorithms, it is hard to gain a constant and exact fusion effect since such fusion algorithms were usually sensitive to the initial value.

In recent years, accompanied by the continuous advancement of neural networks, deep learning has been broadly applied in HSI-MSI fusion tasks. Xie et al. [11] gave a fusion algorithm based on deep learning. Dian et al. [12] proposed a method for HSI-MSI fusion by combining a

Manuscript received 16 July 2022; revised 26 August 2022, 22 September 2022, and 20 October 2022; accepted 21 October 2022. Date of publication 26 October 2022; date of current version 10 November 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 62172139, Grant U1936204, and Grant U1803119; in part by National Key Research and Development Plan under Grant 2020AAA0106800; in part by the Natural Science Foundation of Hebei Province under Grant F2022201055; in part by the Science Research Project of Hebei Province under Grant BJ2020030; in part by China Postdoctoral under Grant 2022M713361; in part by the Open Project Program of the National Laboratory of Pattern Recognition (NLPR) under Grant 202200007; in part by the Natural Science Interdisciplinary Research Program of Hebei University under Grant DXK202102; in part by the Research Project of Hebei University Intelligent Financial Application Technology (HUIFAT) Research and Development Center under Grant XGZJ2022022; and in part by the High-Performance Computing Center of Hebei University. *(Corresponding authors: Siyuan Liu; Weiming Hu.)*

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Digital Object Identifier 10.1109/TGRS.2022.3217168

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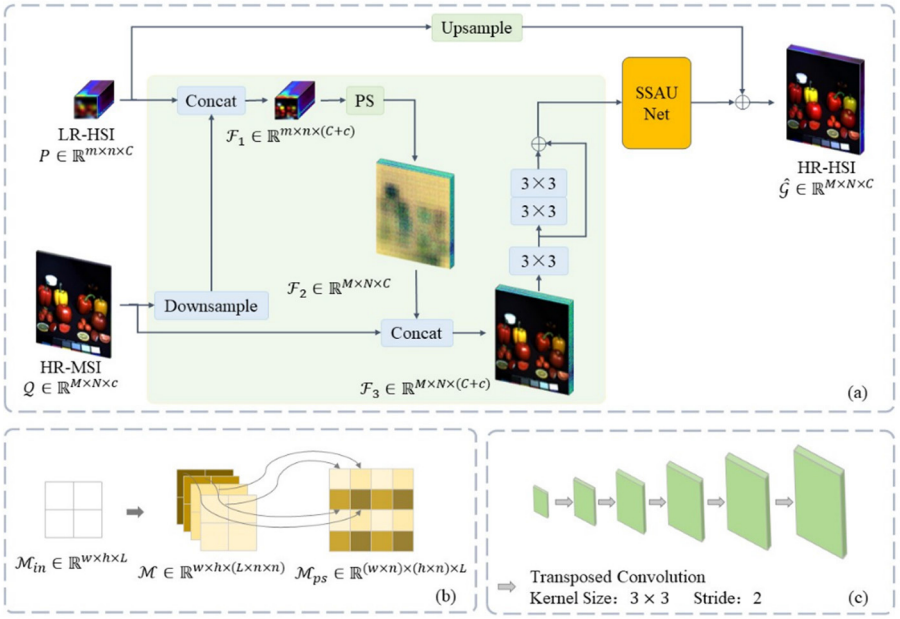


Fig. 1. Overall network structure of our method: (a) overall network framework; (b) PS module; and (c) upsampling module.

convolutional neural network denoiser and subspace representation. Bandara et al. [13] proposed a hyperspectral sharpening method based on improved deep image prior and residual reconstruction, which achieved satisfactory results by strengthening the learning of advanced features. Lee et al. [14] proposed a shift-invariant pan-sharpening network for satellite image moving target alignment. Combined with an attention module, Liu et al. [15] proposed a HSI-MSI fusion algorithm based on an unsupervised multiattention-guided network (UMAG) that does not need training data. The proposed fusion model without any training can generate HR-HSI directly from a pair of input images. Xie et al. [16] proposed an interpretable deep network named MHF-Net for HSI-MSI fusion. MHF-Net not only can intuitively observe the transmission of network internal features but also has good generalization ability. It achieved a good fusion effect.

Even though all the above deep learning-based fusion methods have attained satisfactory fusion effects, there are still several problems. First, because the spectral bands of MSI and HSI do not coincide, many networks designed in deep learning-based HSI-MSI fusion algorithms cannot fully retain the spectral and spatial information in the input image pairs. In the proposed method, the image enhancement module fully retains the nonoverlapping information in MSI and HSI through a series of upsampling, downsampling, and cascading operations on the input source images. Second, most deep learning-based fusion algorithms do not fully exploit the spatial features of images at different scales, which makes the effect of image fusion need to be improved. Therefore, we use a U-shaped structure to construct the network, which fully preserves spatial features of images at different scales. Moreover, we construct a spectral–spatial attention module (SSAM) by combining two attention modules, namely, the coordinate attention (CA) module and the efficient pyramid split attention (EPSA) module, which can fully capture the effective spatial and spectral information. Combining image enhancement modules, U-shaped structure, and SSAM, we have constructed a new network, as shown in Fig. 1. The new network heightens the details in spatial dimensions and spectral dimensions of the inputs, and fully extracts the deep and shallow features of the image.

In conclusion, the contributions of this article are as follows.

1) We propose a spectral–spatial attention-based U-Net (SSAU-Net) for HSI-MSI image fusion, which adequately conserves the spatial and spectral information of the source images.
2) An SSAM is constructed by using a CA module and an EPSA module. This module not only extracts the exact position information of features but also efficiently extracts the channel attention information of features along the spectral dimension.
3) By comparing with the existing advanced methods on different datasets, we find that the proposed algorithm can better reconstruct the details and texture features of fused images. At the same time, the validity and reliability of the proposed algorithm are verified.

The rest of this article is summarized as follows. Section II introduces the related work of HSI-MSI fusion. Section III introduces the proposed network in detail. In Section IV, a large number of experiments are presented to verify the effectiveness of the proposed method. Section V summarizes and anticipates the work of this article.

II. Related Work

At present, we can divide HSI-MSI fusion methods into four categories according to their characteristics, including extended fusion methods based on pan-sharpening, HSI-MSI fusion methods based on matrix decomposition, HSI-MSI fusion methods based on tensor factorization, and HSI-MSI fusion methods based on deep learning. Panchromatic sharpening methods are mainly

divided into three categories: multiresolution analysis, spectral reconstruction, and component replacement. The multiresolution analysis mainly uses a multiscale geometric transform to derive spatial details of MSI and infuse them into HSI. It can preserve the spectral characteristics of HSI well. For example, Liu [17] proposed an image fusion method based on dual-scale decomposition. However, multiresolution analysis has a limited effect on improving the spatial resolution and is time-consuming. In spectral reconstruction, MSI is regarded as the ideal degraded image of HR-HSI, and then, HR-HSI is reconstructed by a traditional random process by using spatial details of a panchromatic image. It is difficult to construct a reconstruction model due to lack ground-truth (GT) image which remains limited room for improvement. Component replacement converts HSI to different color spaces or principal components, and finally, HR-HSI is obtained by corresponding inverse transform after the replacement of main components with MSI. For example, Aiazzi et al. [18] improved the spectral quality of outputs by using the multiple regression method based on the well-established component substitution. It boosted the spatial resolution of MSIs but also caused some spectral distortion. The degradation of spatial resolution from high to low can be regarded as the process of pixel mixing, while the reverse process of pixel mixing, namely, unmixing, can be regarded as image fusion. Hence, MSI fusion based on matrix decomposition can be adopted. The HSI-MSI fusion method based on matrix decomposition has achieved great success, but it is difficult to represent multimodal data by matrix effectively. Consequently, this kind of algorithm has problems such as a large computation and a slow convergence speed. Tensor representation [19], [20], [21], [22] has achieved a great success in HSI-MSI fusion. For example, Li et al. [19] proposed a fusion method based on coupled sparse tensor factorization, which treated HR-HSI as a 3-D tensor. Based on the Tucker decomposition, the fusion problem was defined as the appraisal of three-mode dictionaries and core tensors. Experiments showed that this algorithm carries out considerable image fusion results.

With the development of deep learning, the HSI-MSI fusion method based on deep learning shows excellent performance. For instance, Yang et al. [23] proposed a deep network-based pan-sharpening method, in which an upsampled MSI was added to the output to retain spectral information, while training was performed in the high-pass filter domain to preserve the image spatial structure. However, despite the pan-sharpening image fusion method is relatively simple, the input information is limited, and the fusion effect needs to be improved. Therefore, the general sharpening method is extended to the HSI-MSI fusion method gradually. Hu et al. [24] gave a method of deep spatial–spectral attention network for HSI-MSI fusion, in which spatial attention and channel attention were added to modify spectral and spatial information to achieve HSI-MSI fusion. Hu et al. [24] proposed an HSI pan-sharpening method based on the convolutional neural network, which directly learned the prior information of images through residual learning and achieved good reconstruction effects. Bandara and Patel [26] proposed a texture and spectral feature fusion transformer for pan-sharpening, which improved the spatial and spectral quality of fusion results by learning the spatial correlation and long-distance details of cross features.

In deep learning, U-Net has widespread application in the image processing field. For example, Liu et al. [27] proposed an emotion recognition algorithm based on a dynamic network by combining dynamic entropy features. Li et al. [28] put forward a pancreas segmentation algorithm based on the combination of embedded dual-context features and a multiattention mechanism. Similarly, U-Net has been broadly used in the field of image fusion. For instance, a global feature encoding U-Net proposed by Xiao et al. [29] realized multifocus image fusion. It regarded the focus image generation as a global two-class segmentation mission by U-Net. Liu et al. [30] applied U-Net to HSI classification in combination with superpixels. U-Net fused the features from coarse to fine and obtained good classification results. Qu et al. [31] gave an HSI-MSI fusion algorithm based on an unsupervised bidirectional Dirichlet network. This algorithm adopted a shared encoder–decoder structure to attain spectral and spatial information of different dimensional modes without source image alignment or training which achieved a good fusion effect. Therefore, we also adopt U-Net as the backbone network of the HSI-MSI fusion algorithm in this article. Humans tend not to pay attention to the whole scene when they perceive things but to pay attention selectively. Inspired by this, scholars have brought attention mechanisms to deep neural networks. The attention mechanism effectively focuses on important information and weakens unimportant information. The typical attention modules include the bottleneck attention module and the convolutional block attention module, which attempt to reduce the channel dimension of the input tensor. Then, the spatial attention weight is calculated by convolution [32]. However, the traditional attention module cannot pay attention to spatial and spectral dimensions at the same time, so we will improve it in this study.

III. Proposed Method

A. Fusion Model

In general, the spatial degradation and spectral degradation of HR-HSI are carried out to obtain LR-HSI and HR-MSI in HSI-MSI fusion methods. In the following equations, HR-HSI is represented by $\mathcal{G} \in \mathbb{R}^{M\times N\times C}$, LR-HSI and HR-MSI are represented by $P \in \mathbb{R}^{m\times n\times C}$ and $Q \in \mathbb{R}^{M\times N\times c}$, respectively, where M and m represent the width of the image ($M \gg m$), N and n represent the height of the image ($N \gg n$), and C and c represent the spectral band number of the image ($C \gg c$). LR-HSI and HR-MSI observation models can be usually indicated as follows:

$$P = SB\mathcal{G}, \quad Q = \mathcal{G}\mathcal{R} \tag{1}$$

where B is the representation of a fuzzy matrix, S represents the subsampling matrix, and $\mathcal{R}$ indicates the spectral response matrix.

It is an ill-posed problem to predict HR-HSI by LR-HSI and HR-MSI. Most existing methods estimate HR-HSI by minimizing the error of image reconstruction. In order to

obtain HR-HSI better and more accurately, the regularization term is usually applied to minimize the error of reconstruction by using image prior, that is,

$$\min_{\mathcal{G}} \mathcal{L} = \lambda_1 f_1(\mathcal{G}, P) + \lambda_2 f_2(\mathcal{G}, Q) + T(\mathcal{G}) \tag{2}$$

where λ_1 and λ_2 are the two regularization parameters, and f_1 and f_2 are used to constrain spatial consistency and spectral consistency, respectively. The regularization term based on image prior is represented by T. In general, f_1 and f_2 can be calculated by the following equations, that is,

$$f_1(\mathcal{G}, P) = ||P - SB\mathcal{G}||_F^2 \tag{3}$$

$$f_2(\mathcal{G}, Q) = ||Q - \mathcal{G}\mathcal{R}||_F^2 \tag{4}$$

where $\|X\|_F^2 = (\sum\sum x_{ij}^2)^{1/2}$ is the Frobenius norm.

The HR-HSI obtained by HSI-MSI fusion based on the deep neural network can be seen as the training of the deep neural network. The mapping relationship between input image pairs and output results is obtained by training, which is the process of optimizing the parameters of the deep learning model. We use G_Θ to represent the deep neural network and Θ to represent all parameters of the network. P and Q represent the input LR-HSI and HR-MSI, respectively. $\mathcal{G}$ represents the real HR-HSI. This process can be expressed as

$$\min_{\Theta} \mathcal{L}(G_\Theta(P, Q), \mathcal{G}). \tag{5}$$

In the HSI fusion method based on the deep neural network, the prior knowledge of HSI is contained in the parameters of the network. The network architecture proposed in this article is introduced in the following.

B. Fusion Algorithm

Fig. 1 shows that the proposed algorithm consists of an image enhancement module and SSAU-Net. First, the input LR-HSI and HR-MSI images are enhanced, and a preliminary feature map is obtained. The image enhancement module is shown in the green part of Fig. 1(a). Then, we use SSAU-Net to enhance the features of different scales in the initial feature map and maintain the spectral and spatial features of the source images. The proposed algorithm adds a skip connection so that the network retains more original information from the input LR-HSI. The final HR-HSI estimation (named $\hat{\mathcal{G}}$) can be obtained by elementwise adding the upsampling result of the input LR-HSI to the feature image output by the SSAM.

Different from common linear interpolation upsampling methods, transposed convolution is used to upsample LR-HSI in our method, which can be seen in Fig. 1(c). Common linear interpolation upsampling methods include bilinear interpolation, nearest neighbor interpolation, and bicubic interpolation, among which the bicubic interpolation is better than the other two. The interpolation operation estimates the unknown pixels through the known pixels. Nevertheless, it does not improve the upsampling results with the changes in the network and has poor adaptability. When upsampling LR-HSI by transposed convolution, upsampling parameters suitable for different networks can be obtained through network training, which is beneficial to boosting the overall performance of the network.

1) Image Enhancement Module: The method of fusing HR-MSI with LR-HSI to obtain HR-HSI is called the HSI-MSI fusion algorithm. In general, the purpose of HSI-MSI fusion is to retain as much spectral and spatial information in the source image as possible. However, the spatial details of LR-HSI and important spectral information of HR-MSI are often ignored, resulting in a certain information loss. Therefore, we first enhance the spectral information in LR-HSI and spatial information in HR-MSI with an image enhancement module.

The image enhancement module is shown in green in Fig. 1(a). Let Q represents HR-MSI and P represents LR-HSI. First, Q is downsampled to make it consistent with P in the spatial dimension. The result of downsampling is cascaded with P to obtain $\mathcal{F}_1 \in \mathbb{R}^{m\times n\times(C+c)}$ that contains spectral details of LR-HSI and HR-MSI meanwhile. Specifically, the downsampled HR-MSI is matrix sliced along the spectral channel to obtain three matrix slices, and then, the three matrix slices are inserted into the front, middle, and tail of LR-HSI in a cascade manner. We insert different bands of HR-MSI into their corresponding intervals to obtain $\mathcal{F}_1$. Taking the Columbia Computer Vision Laboratory (CAVE) dataset as an example, the LR-HSI of the CAVE dataset has 31 bands, covering the wavelength range of 400–700 nm at 10-nm wavelength intervals. We divide the 31 wavebands into two parts (the first part is the first 15 bands, and the second part is the last 16 bands), which are distributed at 400–540 and 550–700 nm, respectively, and expressed as H_1 and H_2. The wavelengths of the three bands of downsampled HR-MSI are roughly distributed in the range of 420–470, 500–570, and 630–780 nm. The three bands are, respectively, represented as M_1, M_2, and M_3. We cascade LR-HSI and downsampled HR-MSI in the way of $[M_1, H_1, M_2, H_1, M_3]$. Then, $\mathcal{F}_1$ is input into the pixelshuffle (PS) module [Fig. 1(b)] to obtain an image $\mathcal{F}_2 \in \mathbb{R}^{M\times N\times C}$ with the same spatial resolution as Q. Not dropping the spatial details of the two input source images, $\mathcal{F}_2$ is cascaded with the input Q to obtain the image $\mathcal{F}_3 \in \mathbb{R}^{M\times N\times(C+c)}$. At this time, $\mathcal{F}_3$ has the spatial information and spectral information from P and Q.

Images may inevitably lose shallow information after a deep network. Therefore, in order to enhance shallow information and increase the learning ability of the image enhancement module, ResNet is added to the image enhancement module. The feature map $\mathcal{F}_3$ is input into ResNet to obtain the initial feature map. ResNet consists of three convolution kernels with the size of 3×3 and the step size of 1. For extracting features in different scales of feature images and fully retaining multiscale features, feature images are input into the SSAU-Net for multiscale feature refinement and retention.

2) Spectral–Spatial Attention-Based U-Net: The convolutional layer learns features of different depth levels through the convolution kernel. Feature maps of different depth levels contain different information. The convolutional neural network captures more shallow texture features of the image. However, with the increasing convolution layer, the network captures more edge features of the image. Texture features reflect the parts of the surface that change slowly and periodically, while edge features reflect the parts of the image where pixel changes are not continuous. Taking flowers in the CAVE dataset as

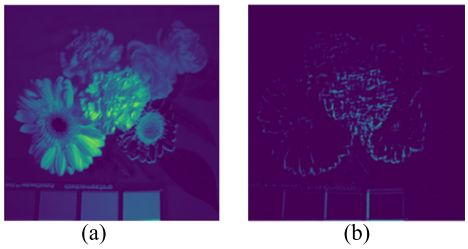


Fig. 2. Deep and shallow features of flowers in the CAVE dataset: (a) texture features extracted from two convolution layers and (b) edge features extracted from eight convolution layers.

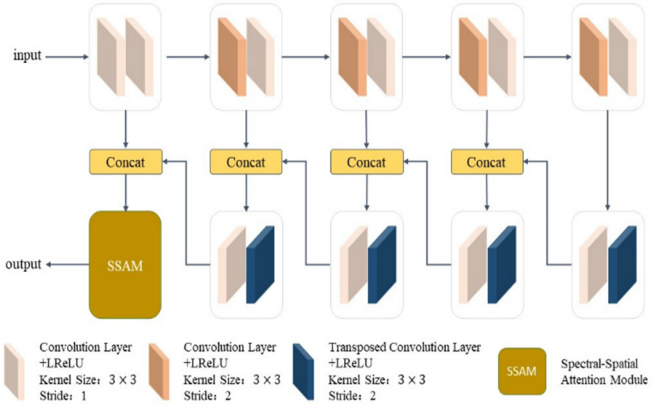


Fig. 3. Structure of SSAU-Net.

an illustration, the differences in features of flowers extracted from different convolutional layers are shown in Fig. 2.

HR-HSI is usually used to complete target recognition, detection, and tracking tasks, while the shallow texture features and deep edge features of images affect the quality of these tasks. However, many methods based on deep learning cannot fully preserve these two features. In this article, we designed SSAU-Net, shown in Fig. 3. SSAU-Net fully preserves the shallow and deep features of the image by cascading feature maps of different depths.

Meanwhile, to enhance spectral and spatial useful information, we design the SSAM and embed it into the U-Net, as shown in Fig. 4. Fig. 4 shows that the SSAM module consists of convolution, CA [33], and EPSA [34]. The feature map input to SSAM is first compressed through a convolutional layer to obtain the same dimension as the output. Then, the CA module is used to encode the position information of the feature map in order to realize spatial attention. The CA module disintegrates the channel attention into two 1-D feature coding operations. Then, it combines features along with two spatial directions separately so as to effectively capture the input location information. The CA module operates layer by layer along the spectral channel, and there is no information interaction of the spectral channel in this process. Therefore, it can only locate important spatial information and cannot reflect the primary and secondary relationship between spectral channels. Finally, we achieved spectral attention by weighting the channels with the EPSA module. It effectively derives the multiscale spatial details of the input feature map at a finer granularity level and simultaneously calibrates the attention vector of the channel dimension. Spatial and spectral attention inevitably weakens the secondary information, and

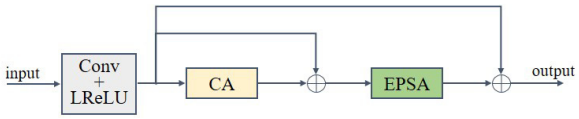


Fig. 4. SSAM.

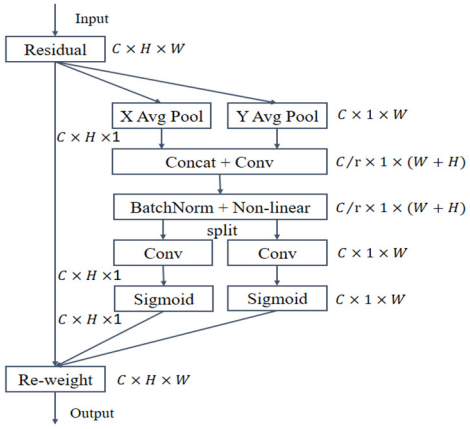


Fig. 5. Structure of CA module. W, H, and C represent the width, height, and number of channels of the feature map, respectively.

the weakening of the secondary information also affects the subsequent application of HSIs. Therefore, jump connections are added in SSAM to prevent the loss of spatial and spectral information. Global pooling operation is commonly used to obtain the attention weight. But it is challenging to retain the location information of feature maps. For accurately capturing the location information of the input feature map, the CA module adopted in this article divides the global pooling into one-to-one feature coding operations. The structure of CA is displayed in Fig. 5.

Each channel of the feature map F can be coded by the CA module along with horizontal and vertical coordinates, respectively. The horizontal coding process of the feature map at the height h of the channel c is indicated as

$$z_c^h(h) = \frac{1}{W} \sum_{0 \le i \le W} F_c(h, i) \quad 0 \le i \le W. \tag{6}$$

Similarly, the process of vertical coding of the feature map at the width w of the channel c is calculated as follows:

$$z_c^w(h) = \frac{1}{H} \sum_{0 \le j \le H} F_c(j, w) \quad 0 \le j \le H. \tag{7}$$

The above operations can catch the long-term dependence of the feature map in one spatial direction while maintaining accurate position information in the other spatial direction by feature aggregation in two directions, respectively. The feature maps obtained by (6) and (7) are cascaded into the shared convolution layer function with the convolution kernel 1×1. To get the medium feature map $\mathcal{F} \in \mathbb{R}^{C/r \times (H+W)}$ (r represents the compression ratio of the channel) which contains horizontal and vertical spatial information. We use Cat$(\cdot)$ to represent the cascade, Conv$(\cdot)$ to represent the convolution operation, and $\delta(\cdot)$ to represent the nonlinear activation function. Then, we obtain

$$\mathcal{F} = \delta\big(\mathrm{Conv}\big(\mathrm{Cat}\big(z^h, z^w\big)\big)\big). \tag{8}$$

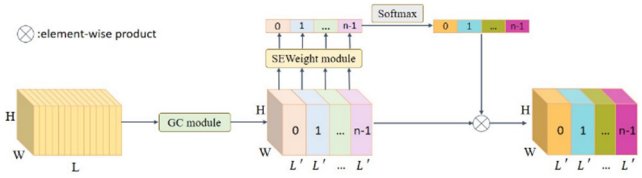


Fig. 6. Structure of EPSA module.

Along the spatial dimension, $\mathcal{F}$ is divided into two independent components $\mathcal{F}^h \in \mathbb{R}^{C/r\times H}$ and $\mathcal{F}^w \in \mathbb{R}^{C/r\times W}$, and then, they are input into two convolution layers Conv_h and Conv_w, respectively, to unify the channel number and input feature map F, which is

$$\mathcal{G}^h = \sigma\left(\text{Conv}_h\left(\mathcal{F}^h\right)\right) \tag{9}$$

$$\mathcal{G}^w = \sigma\left(\text{Conv}_w\left(\mathcal{F}^w\right)\right) \tag{10}$$

where $\sigma(\cdot)$ stands for sigmoid function. Finally, the output y of CA module can be expressed as

$$y_c(i,j) = F_c(i,j) \times \mathcal{G}_c^h(i) \times \mathcal{G}_c^w(j). \tag{11}$$

To enhance the useful spectral details and not interfere with multiscale feature interaction, the EPSA module is integrated into the SSAM to pay spectral attention to the input feature map. The structure of the EPSA module is shown in Fig. 6. The output y of the CA module is divided into S parts along the spectral dimension, denoted by $[y_0, y_1, \ldots, y_{S-1}]$. Different convolution kernel sizes are set for different parts to generate feature maps M of different scales. As the convolution kernel increases, the amount of computation also increases. Therefore, in order to reduce the amount of computation, group convolution is carried out for each part, as shown in Fig. 7.

The relationship between convolution kernel size and group size can be expressed as

$$G = 2^{\frac{K-1}{2}} \tag{12}$$

where K represents the convolution kernel size, and G represents the group size. When the convolution kernel size is set to 3×3, the group size should be set to 1. Then, we use $\text{Conv}(\cdot)$ to represent the convolution operation, y_i to represent the ith part of y, and $K_i = 2\times(i+1)+1$ and G_i to represent the convolution kernel size and group size of the ith part, respectively. Finally, the generating function of the multiscale feature map M_i can be expressed as

$$M_i = \text{Conv}(y_i, (K_i \times K_i, G_i)), \quad i = 0, 1, 2, \ldots, S-1. \tag{13}$$

After multiscale feature processing, the feature map M_i is cascaded to obtain

$$M = \text{Cat}(M_0, M_1, \ldots, M_{S-1}) \tag{14}$$

where $M \in \mathbb{R}^{C\times H\times W}$ stands for multiscale feature map, and $\text{Cat}(\cdot)$ indicates the cascading. The attention weight of different scale features is calculated according to the multiscale feature map, and the calculation process can be expressed as follows:

$$w_i = \text{SEWeight}(M_i) \quad i = 0, 1, 2, \ldots, S-1 \tag{15}$$

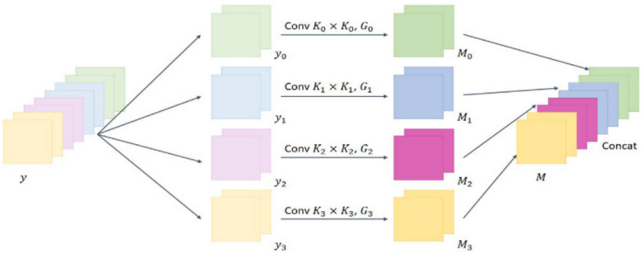


Fig. 7. Group convolution module. K_i and G_i to represent the kernel size and group size of the ith part.

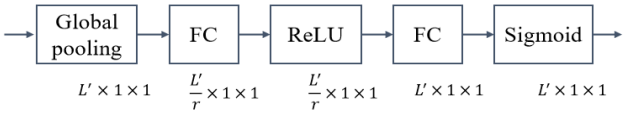


Fig. 8. Structure of SEWeight module.

where w_i represents the attention weight of the ith part and $\text{SEWeight}(\cdot)$ represents the SEWeight module. Fig. 8 shows the structure of the SEWeight module.

As shown in Fig. 8, the SEWeight module first generates channel statistics information through the global average pooling layer. Then, it squeezes and excites channels through two full-connection layers to realize the interaction of high- and low-dimensional channel information and finally assigns weights to channels through the sigmoid layer. Therefore, the complete attention weight is obtained by cascading as

$$w = \text{Cat}(w_0, w_1, \ldots, w_{S-1}) \tag{16}$$

where $\text{Cat}(\cdot)$ indicates the cascading. The weight w was recalibrated by Softmax to obtain the final attention weight $W = \text{Softmax}(w)$. Finally, the recalibrated attention weight is multiplied by the multiscale feature map to obtain the output I. I is calculated as follows:

$$I = M \otimes W \tag{17}$$

where M represents the multiscale feature map, and $\otimes$ represents the elementwise multiplication.

The EPSA module integrates the multiscale spatial information and channel attention weight. It fully extracts the multiscale spatial information and important cross-channel features, which is conducive to realizing HSI-MSI fusion.

3) Loss Function: HR-HSI, LR-HSI, and HR-MSI are represented by $\mathcal{G} \in \mathbb{R}^{M\times N\times C}$, $P \in \mathbb{R}^{m\times n\times C}$, and $Q \in \mathbb{R}^{M\times N\times c}$, respectively, while the fusion image HR-HSI is represented by $\hat{\mathcal{G}}$. Although the commonly used loss function can narrow the gap between the predicted image and the real image, most of them are not designed according to the characteristics of HSIs. In order to conquer this shortcoming, the spectral angle mapping (SAM) [43] constraint is added to the loss function. The loss function is defined as

$$\mathcal{L} = \|\hat{\mathcal{G}} - \mathcal{G}\|_{\text{F}}^2 + \alpha\frac{1}{C}\sum_{j=1}^{C}\arccos\left[\frac{\langle \mathcal{G}, \hat{\mathcal{G}}\rangle}{\|\mathcal{G}^j\|_2\|\hat{\mathcal{G}}^j\|_2}\right]. \tag{18}$$

The first term in (18) can constrain the correlation between fusion result $\hat{\mathcal{G}}$ and GT image $\mathcal{G}$. The second term represents the SAM loss. In the second term, C is the number of spectral pixels, $\mathcal{G}^n$ and $\hat{\mathcal{G}}^n$ represent the jth spectral pixel

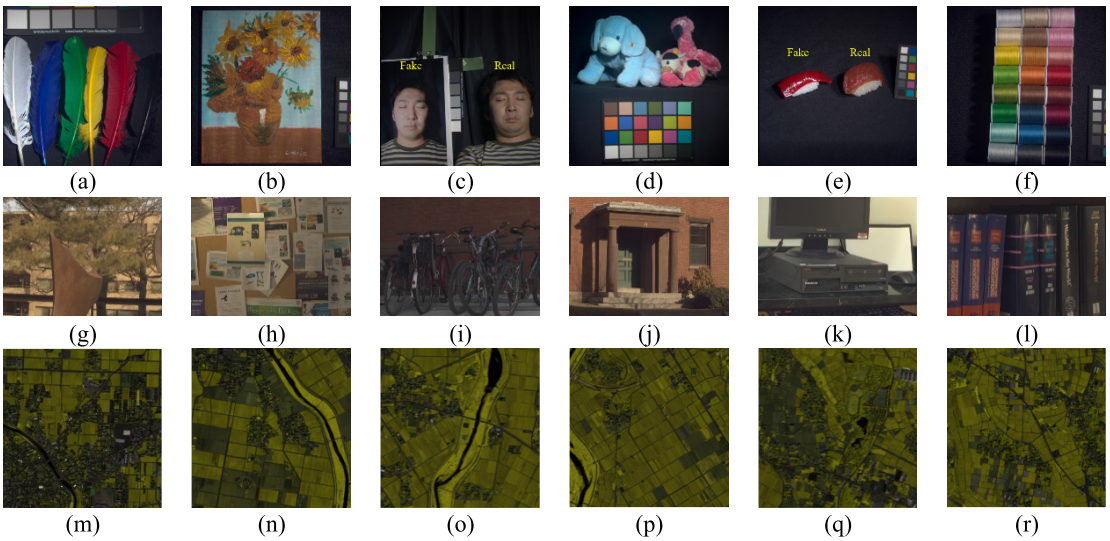


Fig. 9. Examples of false color images from CAVE, Harvard, and Chikusei datasets. The first row is from the CAVE dataset, the second row is from the Harvard dataset, and the third row is from the Chikusei dataset. Of which: (a) feathers; (b) paintings; (c) photographs and faces; (d) plush toys; (e) sushi; (f) coils; (g) imgb7; (h) imgd7; (i) imgf5; (j) imgf7; (k) imgh1; (l) imgh2; (m) example1; (n) example2; (o) example3; (p) example4; (q) example5; and (r) example6.

of the GT image $\mathcal{G}$ and the estimated result $\hat{\mathcal{G}}$, respectively, $\langle \cdot, \cdot \rangle$ represents the inner product of the two vectors, and $\| \cdot \|_2$ represents the l-2 norm. SAM first calculates the cosine between the spectral vectors of $\hat{\mathcal{G}}$ and $\mathcal{G}$ at a certain spectral pixel by $\langle \mathcal{G}, \hat{\mathcal{G}} \rangle / ||\mathcal{G}_2^j|| ||\hat{\mathcal{G}}_2^j||$, then calculates the included angle between the vectors through arccos$[\cdot]$, and finally obtains the average value of the included angles of all the spectral vectors. SAM loss can better constrain the spectral similarity between the predicted image and the real image. We use α as the balance parameter between the two losses and set α to 0.001 in this article.

IV. Experiment

A. Datasets

To validate the availability of the proposed algorithm, we tested our algorithm on the CAVE dataset [35], the Harvard dataset [36], and the Chikusei dataset [37]. We also compared our method with other advanced methods. Fig. 9 shows some images in the CAVE dataset, the Harvard dataset, and the Chikusei dataset. The three datasets are introduced as follows.

The CAVE dataset consists of HSIs from 32 sceneries. Each image contains 31 spectral bands (with 10-nm intervals, covering the wavelength of 400–700-nm range). The spatial resolution is 512 × 512. We randomly selected 20 images as the training set and the residual 12 as the test set. LR-HSI is obtained by spatial degradation of HR-HSI. Specifically, first, pixels are extracted from HR-HSI in the stride of 32 along the two spatial dimensions to obtain the spatially reduced image, and then, the average blur is performed to obtain LR-HSI. We obtained HR-MSI by using the spectral response function of the Nikon D700 camera as a spectral downsampling function. Specifically, the HR-MSI is obtained by multiplying the HR-HSI with the spectral response function with a size of 31 × 3. In the training phase, to increase the number of training samples and reduce the computational cost, we extracted image blocks with the size of 96 × 96 × 31 from HR-HSI as the GT. The corresponding training sample sizes of HR-MSI and LR-HSI were set to 96 × 96 × 3 and 3 × 3 × 31. The sizes of HR-MSI and LR-HSI in the test set were 512 × 512 × 3 and 16 × 16 × 31, respectively.

Fifty indoor and outdoor HSIs were contained in the Harvard dataset. There were 31 spectral bands (with 10-nm intervals, covering the wavelength of 420–720-nm range) in each image. HSIs in the Harvard dataset were cropped to 1024 × 1024 image blocks, and we randomly selected 30 images as the training set and the residual 20 images as the test set. The training sample generation process is the same as the CAVE dataset.

The Chikusei dataset was taken by the Headwall HyperSpec-VNIR-C sensor in Tsukishi, Japan. This dataset contains 128 spectral bands, covering a range of 343–1018 nm. In order to ensure the quality of fusion, 110 spectral bands were retained after discarding the bands polluted by noise. Then, the spectral response function of Worldview-2 was used for spectral subsampling of HR-HSI, and the size of spectral response function was 110 × 8. The number of spectral bands of the sampled HR-MSI image was 8. When cascading in the image enhancement module, eight bands are separated in the form of 2–3–3. The size of the images in this dataset was 2517 × 2335, and the spatial resolution was 2.5 m. The region 1024 × 1536 in the top left corner of the whole image was selected for training, and then, ten nonoverlapping areas with a size of 512 × 512 in the rest of the image were selected for testing.

The method presented in this article was run with Tensorflow 1.14.0 on Python 3.6. Training and testing were performed on Windows 10 with an Intel[1] Xeon Silver 4214R CPU at 2.40 GHz and an NVIDIA GeForce RTX 2080 Ti GPU. We used the adaptive moment estimation algorithm (Adam) optimizer to optimize the network, in which we set the batch size as 64 and the learning rate as 0.0001. The network was optimized for 100 epochs, with 2000 iterations per epoch.

[1]Trademarked.

B. Comparing Algorithms and Evaluation Indexes

In this article, the proposed algorithm was compared with seven kinds of HSI fusion algorithms, including: 1) deep spatio-spectral convolutional neural network-based HSI super-resolution (HSRnet) [24]; 2) hyperspectral and multispectral data fusion via CNMF unmixing [7]; 3) HSI and MSI fusion method by coupled spectral unmixing (CSU) [8]; 4) semiblind HSI and MSI fusion based on nonlocal sparse tensor factorization (NSTF) [38]; 5) low-rank tensor sequence representation-based HSI-MSI fusion (LTTR) [39]; 6) HSI and MSI fusion based on UMAG [15]; and 7) unidirectional total variational HSI-MSI fusion method based on tucker decomposition (UTV) [40].

It is one-sided to gauge the quality of fused images only through subjective visual effects. In order to evaluate the experimental results more comprehensively, eight common evaluation indicators were used in this article, namely, correlation coefficient (CC), the mean of absolute error (MoAE), peak signal-to-noise ratio (PSNR) [41], root-mean-square error (RMSE), structural similarity index measurement (SSIM) [42], SAM [43], universal image quality index (UIQI) [44], and the error of relative dimensionless global in synthesis (ERGAS) [44]. These evaluation indicators are described in detail below. In this article, $\mathcal{G}$ is set as the GT image, $\hat{\mathcal{G}}$ is the estimated HR-HSI image, the spatial range of the image is $M \times N$, and the position of the pixel is represented as (i, j).

CC measures the degree of correlation between two images. The larger the value of CC is, the stronger the correlation may be; otherwise, the lower the value of CC is, the weaker the correlation may be. The formula for CC is defined as (19), shown at the bottom of the page, where $\bar{\mathcal{G}}$ and $\bar{\hat{\mathcal{G}}}$ represent the average value of $\mathcal{G}$ and $\hat{\mathcal{G}}$, respectively. When CC is used for HSI quality evaluation, it is necessary to get the CC value for each spectral band and then take the average value. The closer the CC is to 1, the better the estimated $\hat{\mathcal{G}}$ quality is.

The dissimilarity between the two images can be measured by MoAE. The lower the MoAE value is, the higher the similarity between two images is. This indicates the better quality of the fused result. MoAE is calculated as follows:

$$\text{MoAE} = \frac{1}{MN}\sum_{i=1}^{M}\sum_{j=1}^{N}|\mathcal{G}(i,j) - \hat{\mathcal{G}}(i,j)|. \tag{20}$$

The accuracy of the reconstructed image can be measured by the RMSE. The closer the RMSE value is to 0, the higher accuracy of the output image is. The formula is as follows:

$$\text{RMSE} = \sqrt{\frac{1}{MN}\sum\nolimits_{i=1}^{M}\sum\nolimits_{j=1}^{N}\left|\mathcal{G}(i,j) - \hat{\mathcal{G}}(i,j)\right|}. \tag{21}$$

PSNR is used to calculate the distortion of an image. When the PSNR value is higher, the less distortion there is in the fused result. PSNR can be defined as

$$\text{PSNR} = 10\log_{10}\left(\frac{\text{MAX}}{\text{RMSE}}\right)^2 = 20\log_{10}\frac{\text{MAX}}{\text{RMSE}} \tag{22}$$

where the mean square error is indicated as RMSE, and the maximum value of the image color is indicated as MAX.

ERGAS is used to measure the retention degree of fusion image to GT structure information. The smaller the ERGAS value is, the more fully the structural information is retained. Its definition is as follows:

$$\text{ERGAS}\left(\mathcal{G}, \hat{\mathcal{G}}\right) = \frac{100}{d}\sqrt{\frac{1}{S}\sum\nolimits_{i=1}^{S}\left(\frac{\text{RMSE}\left(\mathcal{G}_i, \hat{\mathcal{G}}_i\right)}{\mu\left(\hat{\mathcal{G}}_i\right)}\right)^2} \tag{23}$$

where $\mu(\cdot)$ represents the mean value of the image, d represents the spatial sampling factor, S represents the spectral subbands number of the image, and $\mathcal{G}_i$ and $\hat{\mathcal{G}}_i$ represent the ith subband of $\mathcal{G}$ and $\hat{\mathcal{G}}$, respectively.

The degree of spectral distortion can be measured by SAM. When the SAM value is smaller, the distortion of the spectrum of the fusion result is smaller. SAM is calculated as follows:

$$\text{SAM}\left(\mathcal{G}, \hat{\mathcal{G}}\right) = \frac{1}{C}\sum_{j=1}^{C}\arccos\frac{\langle \mathcal{G}, \hat{\mathcal{G}}\rangle}{\|\mathcal{G}^j\|_2\|\hat{\mathcal{G}}^j\|_2} \tag{24}$$

where C is the number of spectral pixels, $\mathcal{G}^j$ and $\hat{\mathcal{G}}^j$ represent the jth spectral pixel of the GT image $\mathcal{G}$ and the estimated image $\hat{\mathcal{G}}$, respectively, $\langle\cdot,\cdot\rangle$ represents the inner product of the two vectors, and $\|\cdot\|_2$ represents the l-2 norm function.

SSIM can measure the structural similarity between the fusion result and the true image. SSIM is expressed as

$$\text{SSIM} = \frac{\left(2\mu_{\mathcal{G}}\mu_{\hat{\mathcal{G}}} + c_1\right)\left(\sigma_{\mathcal{G}\hat{\mathcal{G}}} + c_2\right)}{\left(\mu_{\mathcal{G}}^2 + \mu_{\hat{\mathcal{G}}}^2 + c_1\right)\left(\sigma_{\mathcal{G}}^2 + \sigma_{\hat{\mathcal{G}}}^2 + c_2\right)} \tag{25}$$

where $\mu_{\mathcal{G}}$ and $\mu_{\hat{\mathcal{G}}}$ represent the mean of $\mathcal{G}$ and $\hat{\mathcal{G}}$, respectively, $\sigma_{\mathcal{G}}$ and $\sigma_{\hat{\mathcal{G}}}$ represent the variance, and $\sigma_{\mathcal{G}\hat{\mathcal{G}}}$ represents the covariance of $\mathcal{G}$ and $\hat{\mathcal{G}}$. $c_1 = (a_1L)^2$ and $c_2 = (a_2L)^2$ is the constant used to keep constancy, where $a_1 = 0.01$ and $a_2 = 0.03$. L is the dynamic range of pixel values, generally set as 255.

UIQI is used to measure the contrast ratio distortion, luminance distortion, and correlation loss between fusion result and truth image. When the value of UIQI is closer to 1, the reconstructed image quality is more excellent. UIQI is calculated by

$$\text{UIQI}\left(\mathcal{X}, \hat{\mathcal{X}}\right) = \frac{\sigma_{\mathcal{G}\hat{\mathcal{G}}}}{\sigma_{\mathcal{G}}\sigma_{\hat{\mathcal{G}}}}\frac{2\mu_{\mathcal{G}}\mu_{\hat{\mathcal{G}}}}{\mu_{\mathcal{G}}^2 + \mu_{\hat{\mathcal{G}}}^2}\frac{2\sigma_{\mathcal{G}}\sigma_{\hat{\mathcal{G}}}}{\sigma_{\mathcal{G}}^2 + \sigma_{\hat{\mathcal{G}}}^2}. \tag{26}$$

The linear correlation between $\mathcal{G}$ and $\hat{\mathcal{G}}$ can be measured by the first term of (26). The closeness of the average luminance between $\mathcal{G}$ and $\hat{\mathcal{G}}$ is measured by the second term. The third item measures the contrast similarity between images.

$$\text{CC} = \frac{\sum_{i=1}^{M}\sum_{j=1}^{N}\left(\hat{\mathcal{G}}(i,j) - \bar{\hat{\mathcal{G}}}\right)\left(\mathcal{G}(i,j) - \bar{\mathcal{G}}\right)}{\sqrt{\left(\sum_{i=1}^{M}\sum_{j=1}^{N}\left(\hat{\mathcal{G}}(i,j) - \bar{\hat{\mathcal{G}}}\right)^2\right)\left(\sum_{i=1}^{M}\sum_{j=1}^{N}\left(\mathcal{G}(i,j) - \bar{\mathcal{G}}\right)^2\right)}} \tag{19}$$

TABLE I
MEAN VALUES OF OBJECTIVE MEASURES FOR EACH FUSION METHOD ON 12 TEST IMAGES IN THE CAVE DATASET

Method	CC ↑	MoAE ↓	ERGAS ↓	PSNR ↑	RMSE ↓	SAM ↓	SSIM ↑	UIQI ↑
NSTF	0.9914	1.8617	0.5256	39.9223	3.6594	14.7270	0.9639	0.8192
LTTR	0.9958	1.4249	0.4077	42.4375	2.8937	8.5148	0.9761	0.8929
CSU	0.9934	1.3087	0.4381	40.9423	2.8437	7.2137	0.9796	0.8751
CNMF	0.9922	2.1779	0.5754	38.3476	3.9045	7.8502	0.9579	0.8153
UTV	0.9685	5.5658	1.5364	30.5950	9.6685	14.3235	0.8639	0.6515
HSR	0.9913	3.2988	0.7820	35.0627	4.9132	11.0512	0.9286	0.7282
UMAG	0.9945	**1.1767**	0.3613	42.7315	2.4860	6.5141	0.9799	0.8620
OURS	**0.9959**	1.1867	**0.3464**	**43.0417**	**2.4632**	**6.3161**	**0.9874**	**0.9004**

We objectively evaluate the experimental results by these eight evaluation indexes. Among all objective indicators, ↑ in Table I header indicates that a larger objective indicator value is better and ↓ means that a smaller objective indicator value is better. The optimal value is marked in bold, and the suboptimal value is marked in the underscore.

C. Experimental Results on CAVE Dataset

The test set consisted of 12 randomly chosen images from the CAVE dataset. To show the fusion results of various algorithms more objectively, we take the average value of objective evaluation indexes of 12 images, as shown in Table I. According to Table I, the objective evaluation metrics of the proposed algorithm and UMAG are better than other algorithms. All objective metrics of the proposed algorithm are optimal except MoAE. The proposed algorithm is suboptimal only in the MoAE and is just 0.01 lower than UMAG. The SAM value of our algorithm, which is 0.2 lower than the suboptimal one, is significantly better than other algorithms. This is mainly attributed to the addition of SAM constraints in the loss function of the proposed algorithm. The proposed algorithm also performs well in PSNR, RMSE, SSIM, and UIQI, in which PSNR is 0.3 dB higher than the suboptimal value. RMSE is 0.02 lower than the suboptimal value, SSIM is 0.0075 higher than the suboptimal value, and UIQI is 0.0075 higher than the suboptimal value. The good performance of our algorithm is mainly attributed to the cascading of different level feature maps in the constructed network, which facilitates the retention of image structure information. Radar charts were also drawn based on the PSNR values of the 12 test images in the CAVE dataset to show how each algorithm performed on every image in the test set, as shown in Fig. 10. The larger the radius of the radar charts is, the larger the PSNR value of the test image is, the better the fusion effect is. The outer circle numbers represent 12 test images. It shows that our algorithm has obtained the highest PSNR value on most images.

In order to show the effectiveness of different fusion algorithms more intuitively, the fusion results of each fusion algorithm are given as false color images in this article, which can be seen in Figs. 11 and 12. To show the image in detail clearly, we enlarged some details of the image. Fig. 11 shows that the fused images obtained by the proposed algorithm, UMAG, and HSR are closer to the GT image, while the fused images obtained by NSTF, LTTR, CSU, CNMF, and UTV have different degrees of blurring and color distortion. Among

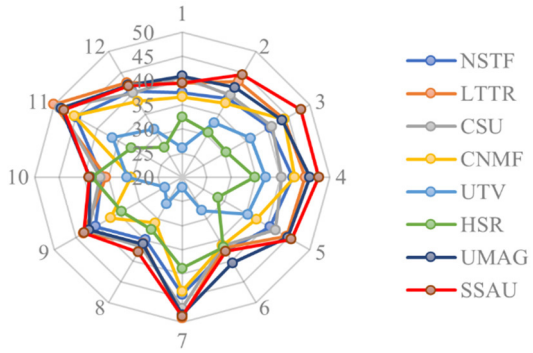


Fig. 10. PSNR values of 12 test images in the CAVE dataset.

them, the fused image obtained by UTV has the most severe border blurring of strawberries, the fused image obtained by LTTR has a significant color change of the green pepper on the left, and the fused image obtained by UTV has severe color distortion.

Fig. 12 shows the magnification of the red box. From the magnified detail part, it can be seen that the red dots on the left side of the red box in the fusion images obtained by NSTF, LTTR, CSU, CNMF, and UTV have different degrees of diminution or disappearance. The fusion images obtained by LTTR and UTV produce serious color distortion. The fused image obtained by HSR produced the block effect, and the fused image boundary of UMAG was blurred. The fused image obtained by our algorithm has the best visual effect. To better observe the results of each fusion algorithm, the error maps of each band of the fused and GT are plotted. The error map of the 26th band of the true and false food images in the CAVE dataset can be seen in Fig. 13, where "Ref" indicates the original image of the corresponding band.

It can be seen from Fig. 13 that NSTF, UMAG, and the proposed algorithm produce the least error. From the two green peppers in the middle of the error map, it can be seen that our algorithm recovers the details of the images better. LTTR, CNMF, and UTV produce serious errors when performing image fusion, and there is a loss of detailed texture. CSU has more difficulty restoring the texture details of the apple and green pepper attachments. HSR produces serious block effect distortion. The 17th band error map of oil painting in the CAVE dataset is given in Fig. 14. Fig. 14 shows that NSTF, CSU, UMAG, and the proposed algorithm have fewer reconstruction errors and obtain better fusion results. Among them, the proposed algorithm better preserves the information of flower stems and petal edges. The fused image of LTTR

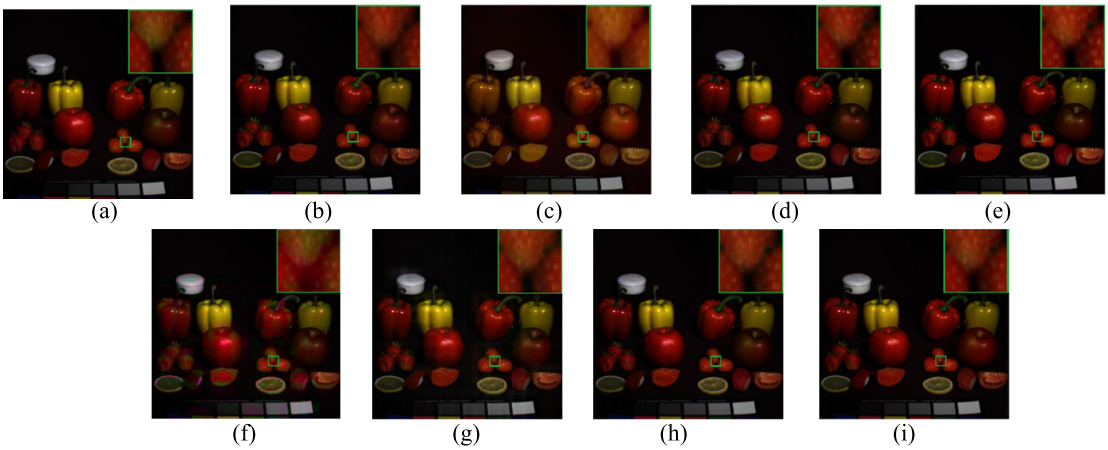


Fig. 11. False color images of various fusion algorithms for fake and real food in the CAVE dataset: (a) GT; (b) NSTF; (c) LTTR; (d) CSU; (e) CNMF; (f) UTV; (g) HSR; (h) UMAG; and (i) OURS.

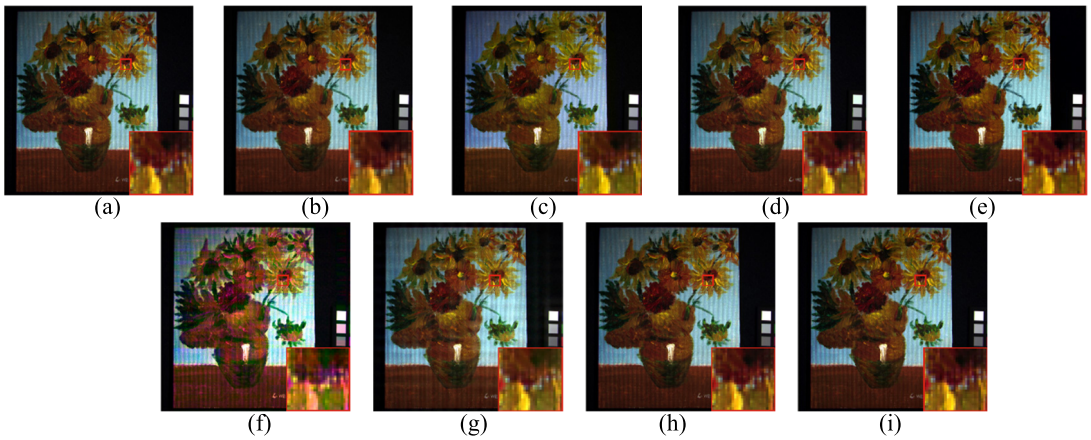


Fig. 12. False color images of various fusion algorithms for oil painting in the CAVE dataset: (a) GT; (b) NSTF; (c) LTTR; (d) CSU; (e) CNMF; (f) UTV; (g) HSR; (h) UMAG; and (i) OURS.

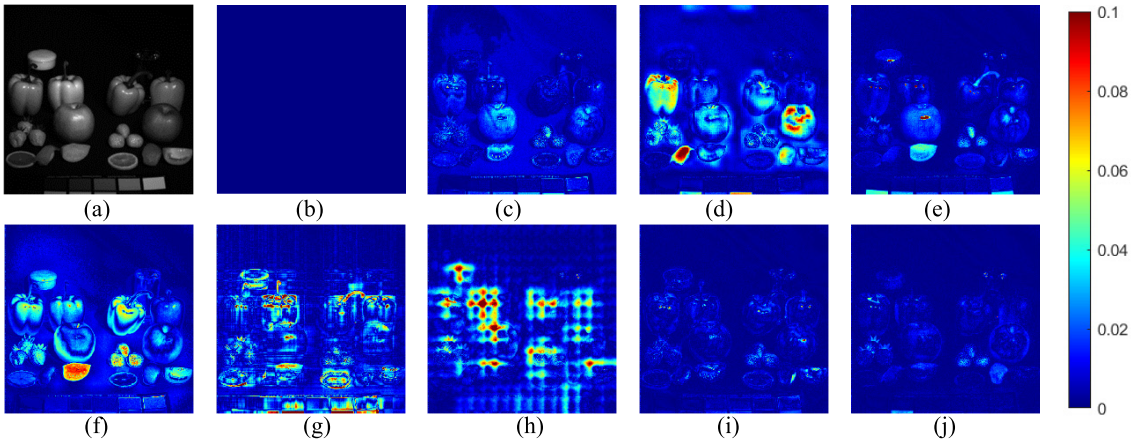


Fig. 13. 26th band error map of each fusion algorithm of fake and real food in the CAVE dataset: (a) Ref; (b) GT; (c) NSTF; (d) LTTR; (e) CSU; (f) CNMF; (g) UTV; (h) HSR; (i) UMAG; and (j) OURS.

has more obvious distortion in the parts of the vase and petal. The fused image of CNMF has an error in the texture of the background of the oil painting. UTV produces serious spatial information distortion. HSR produces a serious block effect.

D. Experimental Results on Harvard Dataset

We chose 20 images from the Harvard dataset as the test set. To display the fusion results of various algorithms more objectively, we take the average value of objective evaluation indexes of 20 images, which can be seen in Table II. Compared with the dataset of CAVE, the image reconstruction of the Harvard dataset is less difficult. Table II shows that the objective effect of the proposed algorithm and UMAG is better than other algorithms. The values of all objective metrics of the proposed algorithm are optimal, except for the suboptimal value on ERGAS, which is only 0.0064 higher. This fully illustrates that the proposed algorithm better maintains the spectral

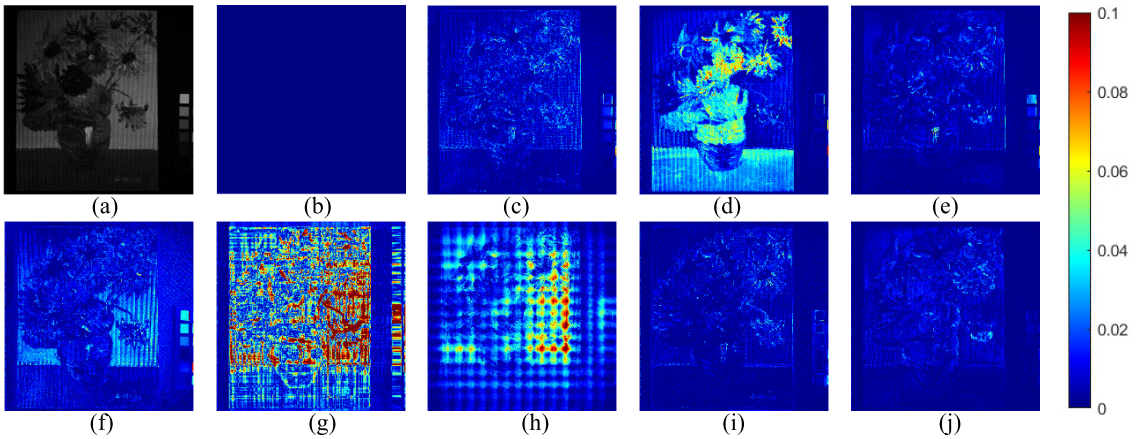


Fig. 14. 17th band error map of each fusion algorithm of oil painting in the CAVE dataset: (a) Ref; (b) GT; (c) NSTF; (d) LTTR; (e) CSU; (f) CNMF; (g) UTV; (h) HSR; (i) UMAG; and (j) OURS.

TABLE II
MEAN VALUES OF OBJECTIVE MEASURES OF EACH FUSION METHOD ON 20 TEST IMAGES IN THE HARVARD DATASET

Method	CC ↑	MoAE ↓	ERGAS ↓	PSNR ↑	RMSE ↓	SAM ↓	SSIM ↑	UIQI ↑
NSTF	0.9888	1.3197	0.3382	43.6359	2.5926	4.9089	0.9735	0.8182
LTTR	0.9908	1.7246	0.3284	42.3607	3.9116	5.8639	0.9649	0.8352
CSU	0.9887	1.0583	0.3369	44.2404	2.1690	3.7212	0.9792	0.8420
CNMF	0.9896	1.3854	0.3466	42.6812	2.5398	3.9000	0.9753	0.8266
UTV	0.9290	3.7923	1.2888	34.7574	6.5734	11.1734	0.8732	0.5927
HSR	0.9884	2.1749	0.4167	39.1433	3.7811	3.9687	0.9722	0.7999
UMAG	0.9910	1.0940	**0.2893**	44.6305	2.0792	3.7590	0.9777	0.8305
OURS	**0.9918**	**1.0205**	0.2957	**44.9230**	**1.9905**	**3.5986**	**0.9803**	**0.8474**

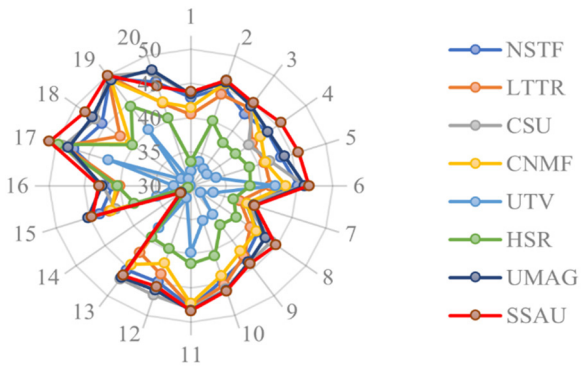


Fig. 15. PSNR values of 20 test images in the Harvard dataset.

and spatial details of the source images. In addition, a radar chart was drawn based on PSNR values of 20 test images in the Harvard dataset, as shown in Fig. 15. It can be seen from Fig. 15 that the proposed method failed to obtain optimal values on only a few images, which once again proves the effectiveness of the proposed method. In order to observe the effectiveness of each fusion algorithm more intuitively, some experimental results of the Harvard dataset are visualized in Figs. 16 and 17. Fig. 16 shows the false color image of imgb8. The detailed part (red box) containing rich scenery information is selected to be enlarged for observation. The fused images obtained by NSTF, CSU, CNMF, UTV, and HSR have color distortion at the location of the green roof, while the fused images obtained by UTV and HSR have partial loss of details of the tree branch in front of the window. The fused image obtained by LTTR produces significant color distortion and artifacts. The proposed algorithm and UMAG have relatively complete preservation of spatial and spectral information. It is obvious that the fusion result obtained by our algorithm is closer to the GT. Fig. 17 shows the false color images of imgh1. Again, it indicates that the fused result obtained by the proposed algorithm has a superior visual effect. In summary, the proposed algorithm obtains satisfactory results.

To better observe the result of each method, we plot the error maps of each band of the fused result and the reference image. The 15th band error map of imgb8 in the Harvard dataset is given in Fig. 18, where "Ref" indicates the original image of the corresponding band. We can see from Fig. 18 that the fused results of CNMF, NSTF, and LTTR have serious distortion at the computer logo. The fused images of UTV and HSR have severe distortion at the wall location. The fused images of CSU and UMAG have smaller distortion around the computer. The proposed algorithm produces the least distortion. Fig. 19 shows the 18th band error map of each fusion algorithm of imgb8, where "Ref" indicates the original image of the corresponding band. It can be seen in Fig. 19 that the fused image of LTTR has an obvious texture loss on the building. The fused image of CNMF has distortion at the margins of the building. The fused result of UTV introduces obvious noise. The fused image of HSR has an obvious block effect on the textured part of the building. The visual results of the proposed algorithm and UMAG are better than other algorithms. It is noteworthy that the error of our algorithm is smaller at the boundary.

E. Experimental Results on Chikusei Dataset

We conducted experiments on the Chikusei dataset to demonstrate the performance of our algorithms for image

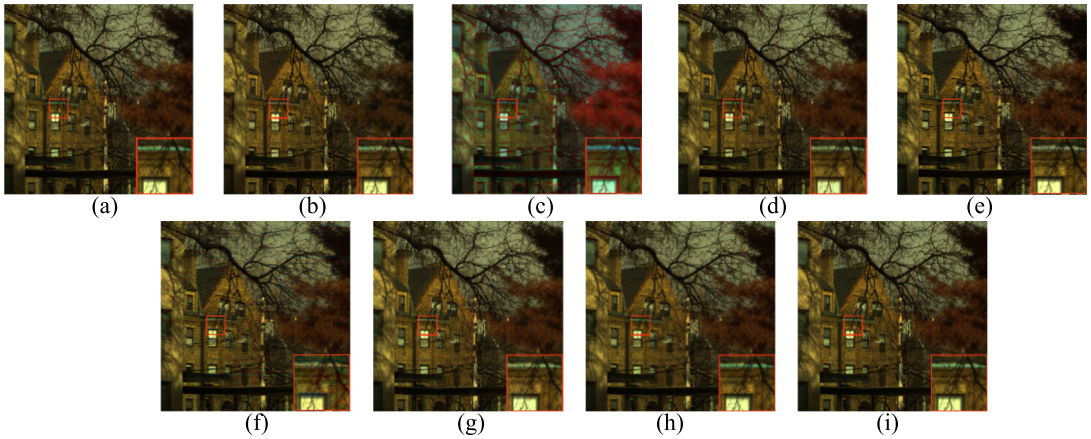


Fig. 16. False color images of various fusion algorithms for imgb8 in the Harvard dataset: (a) GT; (b) NSTF; (c) LTTR; (d) CSU; (e) CNMF; (f) UTV; (g) HSR; (h) UMAG; and (i) OURS.

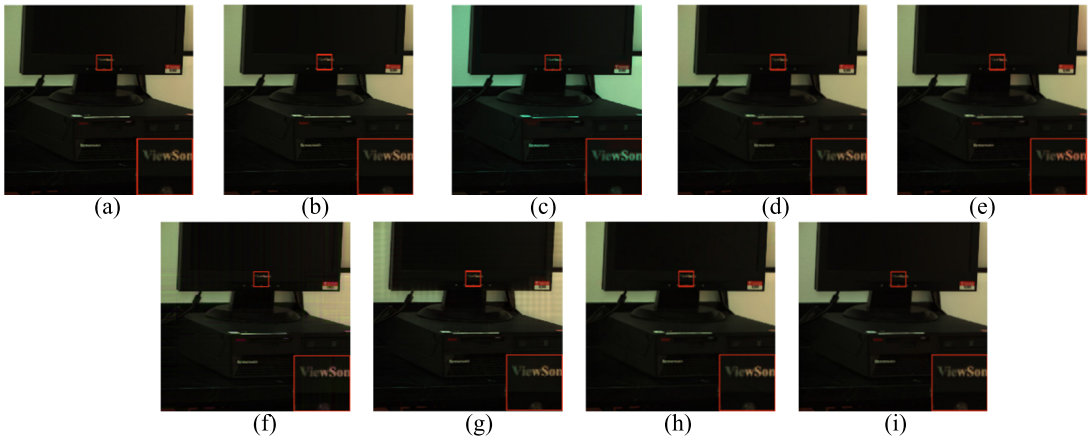


Fig. 17. False color images of various fusion algorithms for imgh1 in the Harvard dataset: (a) GT; (b) NSTF; (c) LTTR; (d) CSU; (e) CNMF; (f) UTV; (g) HSR; (h) UMAG; and (i) OURS.

TABLE III
MEAN VALUES OF OBJECTIVE MEASURES OF EACH FUSION METHOD ON THE CHIKUSEI DATASET

Method	CC ↑	MoAE ↓	ERGAS ↓	PSNR ↑	RMSE ↓	SAM ↓	SSIM ↑	UIQI ↑
NSTF	0.9966	0.2998	0.1933	53.3285	0.7734	1.0023	0.9969	0.9886
LTTR	0.9674	1.9130	0.4410	44.1045	4.4138	5.0682	0.9632	0.9178
CSU	0.9920	0.5513	0.3069	46.8703	1.2942	1.3974	0.9937	0.9789
CNMF	0.9927	0.9373	0.3167	45.2817	1.5634	1.6020	0.9917	0.9729
UTV	0.9333	9.0599	1.2693	33.8536	13.0068	5.8698	0.9034	0.8082
HSR	0.8445	9.3447	1.6545	28.0641	13.8364	6.4607	0.8098	0.6178
UMAG	0.9822	0.9746	0.2732	50.2135	1.6421	1.5338	0.9867	0.9599
OURS	**0.9979**	**0.2903**	**0.1710**	**53.6340**	**0.7055**	**0.9874**	**0.9971**	**0.9897**

fusion on remote sensing datasets. In Table III, we can see the mean values of objective measures of each fusion method on the Chikusei dataset. Table III shows that the objective evaluation indexes of the proposed algorithm are all optimal, and the PSNR value is 0.3 dB higher than the suboptimal value. This indicates that the proposed algorithm is suitable for remote sensing data. In addition, the performance of NSTF on the Chikusei dataset is outstanding. The fusion results of UTV and HSR produce significant spatial distortion and spectral aberration.

At the same time, a radar chart was drawn based on the PSNR values of ten test images in the Chikusei dataset, as shown in Fig. 20. As can be seen from Fig. 20, both the proposed algorithm and UMAG have achieved good results, but the proposed algorithm is more outstanding than UMAG. And the radar chart of SSAU-Net is close to a circle, which indicates that it is stable and has generalization ability in remote sensing data. SSAU-Net does not introduce too many convolution layers in the process of preserving multiscale information of an image by using a U-shaped structure. It only adds the SSAM to the end, which effectively preserves the image's multiscale, spatial, and spectral information and does not produce too much interference information. Meanwhile, the image enhancement module of this article enhances the original information in the early stage, which is also beneficial to improving the performance of the proposed method.

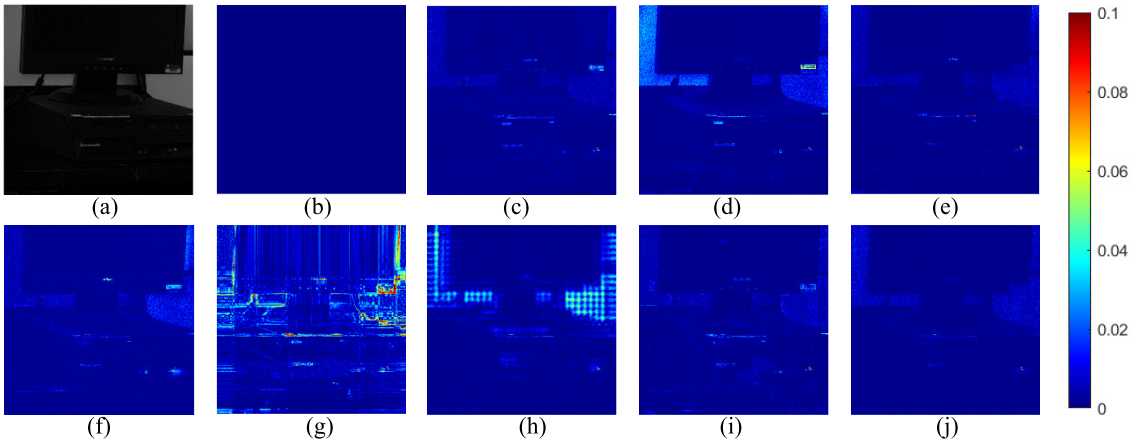


Fig. 18. 15th band error map of each fusion algorithm of imgh1 in the Harvard dataset: (a) Ref; (b) GT; (c) NSTF; (d) LTTR; (e) CSU; (f) CNMF; (g) UTV; (h) HSR; (i) UMAG; and (j) OURS.

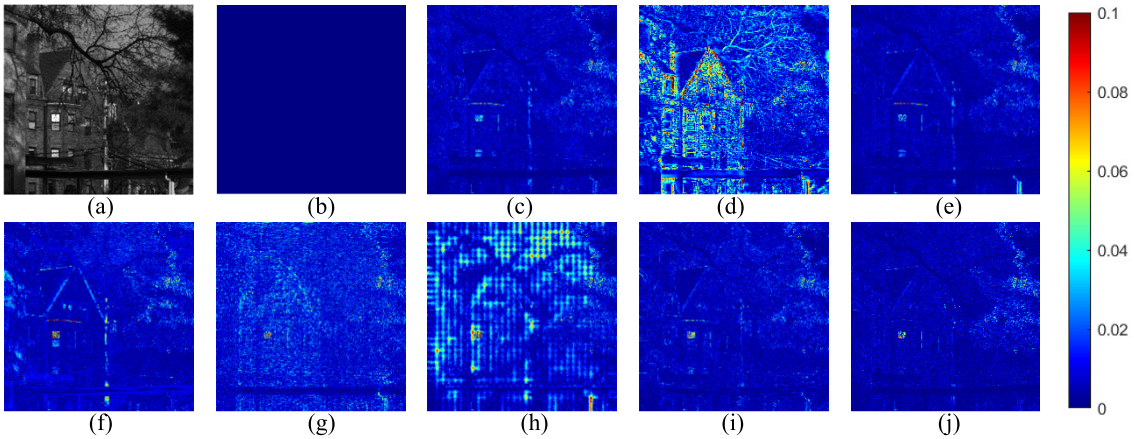


Fig. 19. 18th band error map of each fusion algorithm of imgb8 in the Harvard dataset: (a) Ref; (b) GT; (c) NSTF; (d) LTTR; (e) CSU; (f) CNMF; (g) UTV; (h) HSR; (i) UMAG; and (j) OURS.

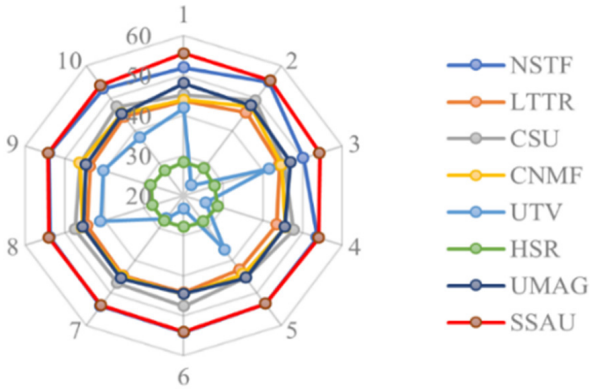


Fig. 20. PSNR values of ten test images in the Chikusei dataset.

Therefore, the proposed method is superior to UMAG. In order to demonstrate the effectiveness of the proposed method more fully, the fusion results on the Chikusei dataset are visualized in this article. Fig. 21 shows the false-color (R-75, G-100, B-36) images of the experimental results obtained by each fusion algorithm on the Chikusei dataset, and the red boxed area is zoomed in. As can be seen from the zoomed area in Fig. 21, the fusion results of CSU, CNMF, and UTV are blurred. The fusion results of HSR show a significant block effect, and the fusion results of LTTR produce color distortion. The fusion images of NSTF, UMAG, and the proposed method obtain better visual results than others.

To better observe the result of each fusion method, the error maps of each band of the fused and GT images are plotted. The 67th band error map in the Chikusei dataset can be seen in Fig. 22, where "Ref" indicates the original image of the corresponding band. The errors in the UTV and HSR are the most obvious, while the NSTF, LTTR, CSU, and CNMF produce more obvious distortions in similar areas. The fusion results of UMAG and our algorithm have the least errors. Especially our algorithm only has insignificant distortion in the lower right corner of the image.

F. Ablation Experiments

The ablation study was carried out to analyze the role of each part of the model, which better verifies the effectiveness of the proposed algorithm. The effects of three components, namely, the CA module, the pyramidal segmentation attention module, and the loss of spectral angular mapping (LSAM) on the fusion results, are verified on the Harvard dataset, as demonstrated in Table IV. In Table IV, w/o-CA indicates that SSAU-Net with the CA module is removed, w/o-EPSA indicates that SSAU-Net with the pyramidal segmentation attention module is removed, and w/o-LSAM indicates that

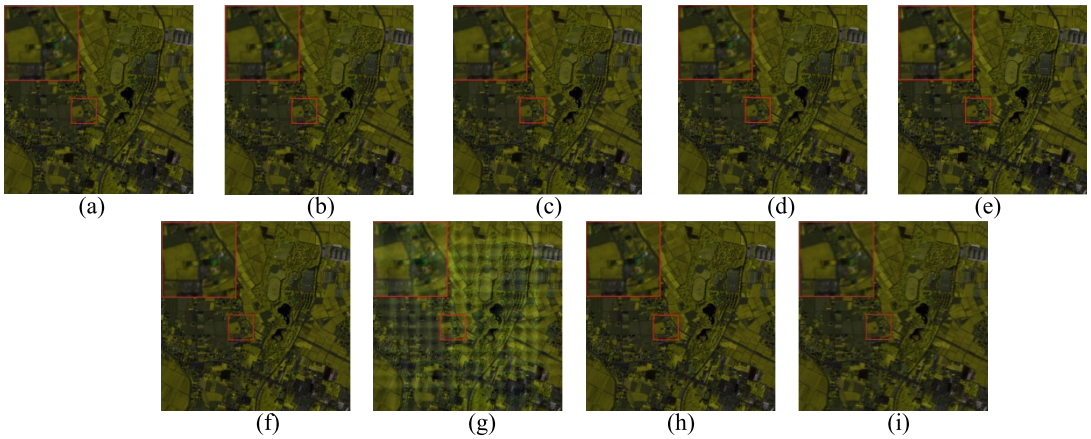


Fig. 21. False color images of various fusion algorithms on Chikusei dataset: (a) GT; (b) NSTF; (c) LTTR; (d) CSU; (e) CNMF; (f) UTV; (g) HSR; (h) UMAG; and (i) OURS.

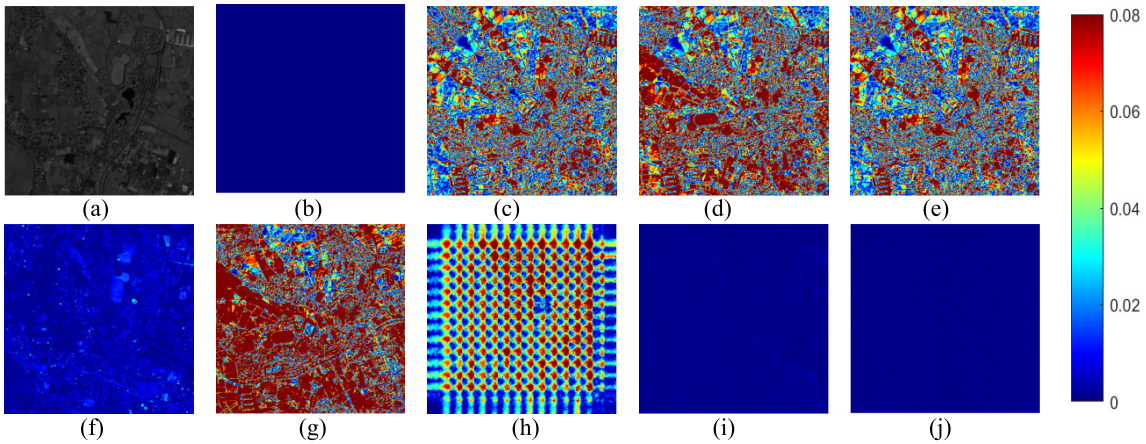


Fig. 22. 67th band error map of each fusion algorithm on Chikusei dataset: (a) Ref; (b) GT; (c) NSTF; (d) LTTR; (e) CSU; (f) CNMF; (g) UTV; (h) HSR; (i) UMAG; and (j) OURS.

TABLE IV

EXPERIMENTAL RESULTS AFTER REMOVING A MODULE IN SSAU-NET ARE COMPARED WITH THOSE OF SSAU-NET

Module	w/o-CA	w/o-EPSA	w/o-LSAM	SSAU
CC ↑	0.9917	0.9908	**0.9918**	**0.9918**
MoAE ↓	1.0498	1.1228	1.0492	**1.0205**
ERGAS ↓	0.2974	0.3156	0.2986	**0.2897**
PSNR ↑	44.9512	44.2563	44.9053	**44.9230**
RMSE ↓	2.0428	2.1229	2.0496	**1.9905**
SAM ↓	3.7018	3.8625	3.6068	**3.5986**
SSIM ↑	0.9800	0.9779	0.9802	**0.9803**
UIQI ↑	0.8461	0.8340	0.8464	**0.8474**

SSAU-Net with the spectral angular mapping loss is removed. Eight objective indicators are used to evaluate the fusion effect, as demonstrated in Table IV, among which the results of the best objective indicators are marked in bold.

We can see in Table IV that the network designed in this article has the best performance, and the network performance decreases to some extent when any part of the network framework is removed. Among them, the performance decreases most significantly when the EPSA block is removed. The PSNR decreases by about 0.7 dB, and the RMSE rises by 0.1324. This indicates that the EPSA has good noise immunity and spatial retention capability. Similarly, when removing the

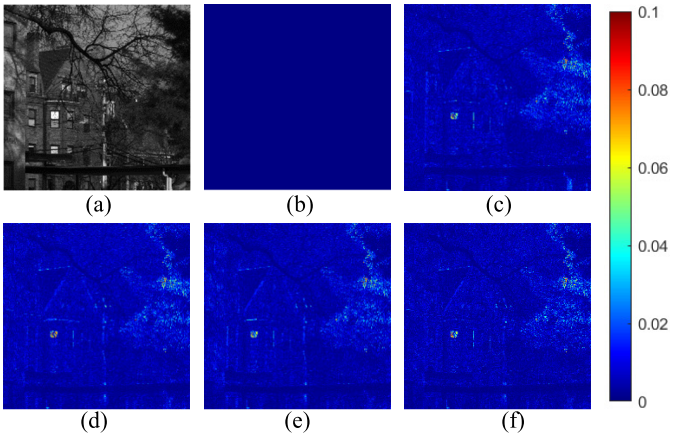


Fig. 23. Error map of the ablation experiment: (a) Ref; (b) GT; (c) w/o-CA; (d) w/o-EPSA; (e) w/o-SAM; and (f) OURS.

CA block, the SAM value of the fusion algorithm is increased by 0.1032, which indicates that the CA block can keep the spectral message of the image well and decrease the spectral distortion of the fused image. With LSAM removed, the SAM of the fusion algorithm rises by 0.0082, which indicates that LSAM can help the proposed algorithm preserve the spectral information of the source images well.

In order to describe the effect of image fusion more intuitively after removing each module, we take imgb8 in the

Harvard dataset as an example and give its error map in the 18th band, as shown in Fig. 23. From Fig. 23, it can be seen that the error of the window on the left building in the fused image becomes larger after removing the CA module or EPSA module. In contrast, the error of the tree shadow on the left building increases significantly after removing the SAM loss. SSAU-Net maintains the edge detail and texture detail of the image well.

V. Conclusion

We construct an SSAU-Net for the HSI-MSI fusion in this article. SSAU-Net first enhances the features of the images by image enhancement to upgrade the stability of the network. Then, the spectral and spatial information about the input image pairs can be effectively maintained by constructing the SSAM. Finally, U-Net acquires the input multiscale features from coarse to fine to produce high-quality fused results. Compared with existing HSI-MSI fusion methods, the proposed algorithm achieves superior fusion results in terms of both visual effects and objective metrics. However, the stability of the proposed algorithm needs to be further tested, and the proposed algorithm will be further tested with other remote sensing datasets.

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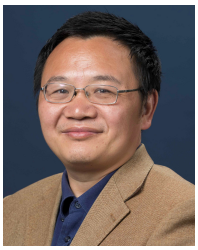
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